CS341 · COMPUTER GRAPHICS

Midterm 461 Practice

20 marks

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01 Questions

Q1 SHORT ANSWER

1. Define computer graphics.

2. What is the difference between absolute and relative coordinate input?

3. A screen has a horizontal length of 10 inches and a vertical height of 5 inches. What is the aspect ratio given that the screen resolution is 1920 × 1080 points?

Q2 TRUE / FALSE

1. Plasma displays are more energy-efficient than LCD displays. ( )
2. M + Y + C = White. ( )
3. A raster scan device scans the screen from vertical to horizontal. ( )
4. Raster scan is more efficient than random scan in drawing lines and shapes. ( )
5. The refresh rate is the measure of how many times the image on a screen is updated per second. ( )
6. Graphic applications are used only for entertainment. ( )

Q3 CHOOSE / COMPLETE

1. The standard design of a video monitor is the _____.
2. Subtractive mixing is the principle of the _____ color model.
3. The frequency at which a picture is redrawn on the screen is referred to as the _____.
4. A goal of computer graphics: _____.
5. The rays emitted by a cathode ray tube are _____.
6. Applications of computer graphics: CAD / Medical visualization / Entertainment / _____.
7. Digital displays use the _____ color model.

Q4 FILL IN THE BLANKS

1. Yellow + Magenta = _____

2. Magenta + Cyan = _____

3. Green + Blue = _____

4. _____ is the maximum number of points/pixels that can be displayed (without overlap) on a CRT.

5. _____ is defined as the time that it takes the emitted light from the screen to decay to one-tenth of its original intensity.

6. _____ is the set of all colors that can be produced from mixing the primary colors.

7. _____ is white pigment added to pure pigment.

8. _____ is the consequence of adding both white and black pigments to pure pigments.

Q5 LIST

1. List four hard-copy devices.

2. List four applications of computer graphics.

3. RGB stands for:

Q6 BRESENHAM

Using Bresenham's algorithm, find the pixels of the line starting at (3, 3) and ending at (10, 7).

k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				

